Efficacy and Safety of Renalix in Chronic kidney disease with albuminuria: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

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Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Renalix (RNX) is a once-daily oral endothelin receptor antagonist under investigation for chronic kidney disease with albuminuria. **Methods:** In this multicentre trial, 1450 adults were randomised 1:1 to Renalix or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was eGFR slope (mL/min/1.73m2 per year). **Results:** The primary endpoint was -1.9 with Renalix versus -3.4 with placebo (p <0.001). Fluid retention occurred in 12.6% of Renalix patients and 6.8% of placebo patients. **Conclusions:** Renalix met its primary endpoint with a manageable safety profile in chronic kidney disease with albuminuria.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of chronic kidney disease with albuminuria and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Renalix	Placebo
Participants (n)	725	725
Mean age (years)	59.5	62.9
Female (%)	55	46
Primary endpoint	-1.9	-3.4
p-value	<0.001	-
Fluid retention (%)	12.6	6.8
Discontinuation due to AE (%)	4.4	1.2

Serious adverse events were reported in 5.2% of the Renalix group and 10.3% of the placebo group. There were 5 deaths in the Renalix arm and 4 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Renalix is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.